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CK1α agonists attenuate medulloblastoma stemness and relapse risk

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ABSTRACT

While outcomes for most children with medulloblastoma (MB) are relatively favorable, those in the Sonic Hedgehog (SHH) subgroup with *Tumor protein P53* (*TP53*) mutations-known as the SHHα subtype-face a much poorer prognosis. SHHα patients relapse more frequently and rapidly, underscoring the need for therapies that prevent recurrence. We recently identified a non-canonical Gli-driven $Sox2^+$ cell population that promotes relapse in SHH MB. However, few Gli-targeting strategies have shown clinical promise to date. One translational Gli inhibitor is pyrvinium, an FDA-approved compound known to destabilize Gli through increasing Casein kinase 1α (CK1α) activity. In this study, we tested whether pyrvinium and a brain-permeable derivative, SSTC3, affect stemness and relapse risk in mouse and human-derived SHHα MB models. We found that pyrvinium suppresses the Gli-driven proliferation of $Sox2^+$ cells. Unlike other SHH/Gli-targeting approaches, pyrvinium also impaired MB self-renewal by depleting Cluster of Differentiation 15 $(CD15)^+$ cells. Mechanistic studies revealed that $CD15^+$ cell self-renewal is WNT-dependent and driven by the loss of p53/*microRNA-34a*-mediated repression of WNT signaling. Remarkably, pyrvinium and SSTC3 reduced $Sox2^+$, $CD15^+$, and dual Sox2/CD15-labeled populations in mouse and patient-derived SHHα models. Consistent with their ability to diminish tumor stemness, pyrvinium also impaired primary and secondary tumor engraftment. Our findings show that CK1α agonists regulate stemness in SHHα MB, establishing CK1α as a therapeutically relevant vulnerability. While pyrvinium itself is not an ideal clinical candidate, these data support the development of second-generation brain-penetrant CK1α-targeting derivatives.

INTRODUCTION

Brain tumors are the leading cause of cancer-related death in children, with medulloblastoma (MB) being the most common malignant form(1). From the four major molecular subgroups, the Sonic Hedgehog (SHH) subgroup accounts for approximately 30% of cases(2). Clinical outcomes in this subgroup are strongly influenced by co-occurring genetic alterations. In particular, *Tumor protein P53* (*TP53*) mutations, present in ~7-30% of cases at diagnosis(3-5) and ~30% at recurrence(4), are associated with increased relapse risk and shorter time to relapse(3), and thus these tumors are classified as very high-risk(6). Given their dismal outcomes, there is an urgent need to understand the mechanisms driving relapse in *TP53*-mutant SHH MB, also known as SHHα(7).

Emerging evidence suggests that a rare population of MB progenitor cells (MPCs), often marked by the stemness factor SRY-box transcription factor 2 (Sox2), may contribute to disease recurrence(8). Our previous work in SHH MB showed that the expansion of $Sox2^+$ cells relies on non-canonical Gli activity, rendering them resistant to Smoothened (Smo) inhibitors in clinical development(9). By contrast, compounds that block Gli-dependent transcription via Bromodomain and extra-terminal (BET) inhibition reduced $Sox2^+$ cells and relapse risk in SHH MB animal models(9). Nevertheless, the toxicity associated with BET inhibitors in clinical settings(10) underscores the need for alternative Gli-targeting approaches to prevent MB recurrence.

One translational approach to target Gli involves activating Casein kinase 1α (CK1α). CK1α is a serine/threonine kinase that phosphorylates Gli, promoting its degradation(11). Small-molecule CK1α agonists, such as the FDA-approved anti-helminthic drug pyrvinium, selectively bind CK1α and function as allosteric activators(12). Through CK1α activation, pyrvinium and a brain-permeable derivative, SSTC3, attenuate SHH-driven MB growth(13, 14). However, CK1α also phosphorylates β-catenin, an essential transcriptional activator in the Wingless-related integration site (WNT) pathway, targeting it for proteasomal degradation(15). Thus, CK1α agonists extend their effect beyond Gli, as both pyrvinium and SSTC3 block WNT signaling and suppress the growth of WNT-driven malignancies(16, 17).

Given their dependence on Gli signaling, we hypothesized that CK1α agonists deplete Sox2^{+} cells implicated in SHHα MB relapse. Here, we show that CK1α agonists target not only Gli-driven Sox2^{+} cells but also a WNT-driven, self-renewing CD15^{+} compartment, thereby limiting the ability of residual disease to re-engraft. Together, these findings support CK1α modulation as a strategy to suppress relapse-driving cell populations in very high-risk SHHα MB.

MATERIAL AND METHODS

Mouse studies

All animal work was conducted in compliance with the NIH Guide for the Care and Use of Laboratory Animals. Experimental protocols received approval from the Institutional Animal Care and Use Committees of the Medical University of South Carolina and the University of Miami. *Ptch1^{tm1Mps}/J* (*Ptch1-LacZ*) and *B6.129S2-Trp53^{tm1Tyj}/J* (*Trp53-KO*) mice (Jackson Laboratory) were crossed to establish a colony. Spontaneous tumors from either female or male mice were expanded and maintained as allografts in 6-10-week-old *CD1-Foxn1^{nu}* male mice (Charles River). For pyrvinium (Enzo or Sigma-Aldrich) treatment, 1×10^{6} viable cells (trypan blue-excluded) were implanted subcutaneously (s.c.) into similar mice. Once tumors reached ~200 mm^3, mice received pyrvinium (0.8 mg/kg, s.c.) near the tumor every other day (q.o.d.). Tumors were harvested 6 hours (h) after the final dose and either fixed or dissociated with Papain (Worthington) or Accutase (Invitrogen) for FACS, sphere formation, or re-engraftment assays.

For flank re-engraftment, indicated viable cells were implanted s.c. into 6-10 weeks old *CD1-Foxn1^{nu}* female or male mice. For orthotopic re-engraftment, 1×10^4 viable cells from pyrvinium-treated tumors were resuspended in 3 µL Neurobasal-A media and implanted into the cerebellum(18). For primary engraftment, MPC-2 cultures were treated with 200 nM pyrvinium for 24h before s.c. implantation.

For SSTC3 (StemSynergy Therapeutics) studies, 6-10 weeks old *CD1-Foxn1^{nu}* and NOD.Cg-Prkdc^{scid} Il2rg^{tm1Wjl}/SzJ male mice were orthotopically implanted with 1×10^5 *Ptch1-LacZ; Trp53-KO* cells or 1×10^6 *TP53*-mutant patient-derived xenograft (PDOX) cells (SJSHHMB-14-7196, courtesy of Dr. Roussel, St. Jude), respectively. Ten- and thirty-day post-implantation, respectively, mice received daily (q.d.) intraperitoneal (i.p.) injections of vehicle or SSTC3 (10 mg/kg) for 3 days. Brains were then harvested for flow cytometry (Accutase) or IHC.

Statistical Analysis

Results represent the mean ± SEM from at least three independent experiments. For BrdU staining, four fields per condition from three experiments were quantified. Multiple group comparisons used one-way ANOVA with post-hoc Dunnett analysis. Two-sample comparisons used one-tailed Student's t-tests. Symptom-free survival was assessed using Log-rank (Mantel-Cox) tests. Tumor engraftment significance was determined using a χ^2 test. Statistical tests were selected as appropriate for each dataset and are indicated in the corresponding figure legends. Data distribution and variance were evaluated before analysis to confirm suitability for parametric testing, and variance was comparable between groups. For in vivo studies, the exact number of biological replicates is indicated in each panel, either by individual dots representing independent tumors or by listing the number. Mice were randomized before dosing and sample size was based on our prior experience with similar models(9, 14, 18). There was no blinding during in vivo experiments. Statistical significance: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$.

Additional Material and Methods are described in Supplemental Methods.

RESULTS

Pyrvinium blocks Gli1-driven Sox2$^+$ cell proliferation

We previously showed that the propagation of MB cultures enriched in Sox2$^+$ cells, herein referred to as MPC cultures, is driven by non-canonical Gli activation downstream of Smo(9). Because CK1α destabilizes Gli(11), we tested the efficacy of pyrvinium, a known CK1α agonist(12), in MPC cultures. Given that most SHH MB relapses are associated with *TP53* mutations(3, 5), we focused on *Trp53* deficient cultures (*Trp53-KO*). According to its mechanism of action, pyrvinium not only reduced *Gli1*-driven promoter activity in a reporter cell line (**Figure 1A**) but also decreased the SHH-driven cell proliferation of the granule layer of the developing cerebellum in organotypic cultures (**Figure 1B**; **Supplemental Figure 1A**). In MPC cultures, nanomolar concentrations of pyrvinium reduced Gli1 levels (**Figure 1C, D**) and the number of Gli1$^+$ cells (**Figure 1E**), thereby attenuating the expression of SHH target genes (**Figure 1F**). Furthermore, pyrvinium reduced the number of viable cells in MPC cultures, with an average EC50 of 17 nM (**Figure 1G**), which is consistent with their dependency on Gli to propagate(9).

Given pyrvinium's ability to reduce the number of viable cells in MPC cultures, we investigated whether this effect was due to decreased proliferation or increased cell death. BrdU incorporation assays suggested that pyrvinium's effects are primarily attributed to reduced proliferation, as evidenced by a drop in BrdU$^+$ cells (**Figure 1H**; **Supplemental Figure 1B**). This was further supported by the reduction in levels of proliferation markers, such as Cyclin-D1 and PCNA

(**Figure 1I, J**), as well as cells labeled for the proliferation marker Ki67 (**Figure 1K**). In contrast, pyrvinium had only minimal effects on cell death, as determined by the number of cleaved Caspase-3-labeled cells (**Figure 1H**; **Supplemental Figure 1B**) and its protein levels (**Figure 1I, J**). Importantly, similar to other Gli inhibitors(9), pyrvinium also reduced the number of $Sox2^+$ cells in MPC cultures (**Figure 1L**), including those $Sox2^+$ cells co-labeled with Gli1 and Ki67 (**Figure 1M**). Together, these data suggest that pyrvinium depletes $Sox2^+$ cells by blocking their Gli-driven proliferation. Supporting a non-canonical Gli activation downstream of Smo, a compound acting on Smo, vismodegib, had little effect on the levels of Gli1 in $Sox2^+$ cell enriched cultures (**Supplemental Figure 1C**). Subsequently, vismodegib neither affected the levels of proliferation and cell death markers (**Supplemental Figure 1D**), or the specific numbers of $Sox2^+$ cells (**Supplemental Figure 1E**) in these cultures.

Pyrvinium attenuates MB self-renewal

MPCs are known to self-renew to maintain tumor stemness(19). Interestingly, while our previous work showed that Gli inhibition by targeting BET proteins reduced MPC culture proliferation(9), the ability of these drugs to block secondary sphere formation, used as a measure of tumor stemness(20), was not tested at that time. To evaluate this, we used self-renewal protocols in which MPC cultures were exposed to drugs of interest for 24h before disaggregating spheres(18). Equal numbers of viable cells were next re-plated and allowed to form new spheres in the absence of drug (**Figure 2A**). These studies revealed that three structurally distinct BET inhibitors (I-BET151, JQ-1, BMS-986158) fail to block the self-renewal of MPC cultures (**Figure 2B**) at concentrations we previously showed to attenuate SHH signaling and reduce MPC viability(9). Similarly, the Smo inhibitor vismodegib failed to impair MPC secondary sphere formation (**Figure 2C**), suggesting that MPC self-renewal is not SHH/Gli-driven. Accordingly, siRNA pools targeting

Gli1 and *Gli2* also failed to inhibit MB self-renewal (**Figure 2D**). In contrast, pyrvinium attenuated secondary sphere formation in MPC cultures, with an average EC50 of 25 nM (**Figure 2E**). Given pyrvinium's effect on MB self-renewal *ex vivo*, we studied whether it similarly affects tumor initiation. MPC cultures were exposed to pyrvinium before implanting limiting dilutions of viable cells into mice. Tumor initiation was observed from the vehicle-treated cultures, even when only 150,000 cells were implanted. However, pyrvinium-treated cells did not form tumors, even when nearly tenfold higher cell numbers were used (**Figure 2F**). Together, these findings suggest that pyrvinium blocks MB self-renewal by targeting a pathway different than SHH/Gli.

Pyrvinium inhibits MB self-renewal by targeting WNT signaling

In addition to regulating Gli stability(11), CK1α is also known to prime β-catenin for degradation(15), thereby suppressing WNT signaling. Moreover, our previous findings indicated that self-renewal in *Trp53-KO* MB is driven by WNT activation(18). Thus, we wondered whether pyrvinium's effects on self-renewal operate through WNT inhibition. Supporting this idea, pyrvinium reduced WNT activity in a TCF/LEF1 reporter assay (**Figure 3A**) and decreased both β-catenin levels (**Figure 3B, C**) and expression of WNT target genes (**Figure 3D**) in MPC cultures. Furthermore, overexpression of a constitutively active β-catenin form (*S33Y* mutant) increased baseline expression of WNT biomarkers (**Figure 3E**) and reduced pyrvinium's efficacy on MPC self-renewal (**Figure 3F**), supporting a WNT-dependent mechanism of action. Further supporting a WNT-mediated self-renewal, vismodegib, which we showed does not alter MPC self-renewal, likewise failed to attenuate WNT signaling (**Supplemental Figure 1F**).

Although our prior work linked WNT activation to p53 loss in MPC cultures(18), the mechanism driving elevated WNT signaling remained unclear. One possible explanation involves *microRNAs* (*miRNAs*), small non-coding RNAs that bind to 3' UTRs of target *mRNAs* and suppress their translation(21). p53 loss has previously been associated with downregulation of *miR-34a*, leading to de-repression of WNT components such as LEF1(22), a key member of the β-catenin transcriptional complex(23). Accordingly, *Trp53-KO* MPC cultures exhibited reduced levels of *miR-34a* compared to wild-type counterparts (**Figure 3G**). Introduction of a *mimic-miR-34a* into *Trp53-KO* MPC cultures decreased LEF1 levels, whereas inhibition of *miR-34a* using an *anti-miR* did not significantly increase LEF1 levels at a similar timepoint (**Figure 3H**). However, an *anti-miR-34a* increased WNT target gene expression in *Trp53-wild-type* MPCs (**Figure 3I**). Further implicating *miR-34a* in MB self-renewal, a *mimic-miR-34a* reduced secondary sphere formation in *Trp53-KO* MPC cultures, while an *anti-miR-34a* had minimal effect (**Figure 3J**). These findings suggest that loss of p53 reduces *miR-34a* levels in MPC cultures, thereby enabling WNT activation (**Figure 3K**).

Pyrvinium depletes a self-renewing CD15$^+$ cell population

The ability of BET inhibitors to deplete Sox2 cells(9) without affecting self-renewal suggests the involvement of an alternative cell pool. To define the self-renewing compartment targeted by pyrvinium, we focused on CD15$^+$ cells, which were previously shown to represent a tumor-initiating population in SHH MB(24-26). CD15$^+$ cells were sorted from *Ptch1-LacZ, Trp53-KO* tumors, using Ter-119 to deplete erythroid lineage cells (**Figure 4A**). Compared to CD15$^-$ cells, CD15$^+$ cells formed more spheres, and pyrvinium reduced these numbers (**Figure 4B**). Furthermore, linking CD15$^+$ self-renewal to WNT signaling, similar to pyrvinium, a compound that blocks β-catenin/LEF1 interaction, PKF115-584(27), impaired sphere formation in similarly sorted cells, while SHH inhibition by vismodegib failed to do so (**Figure 4B**). These findings suggest that pyrvinium targets a WNT-driven CD15$^+$ population. Gene expression analysis further supported this idea, as CD15-sorted cells showed elevated expression of multiple WNT target genes compared to $CD15^-$ cells, while SHH target gene levels were similar (**Figure 4C**). Furthermore, treatment of MPC cultures with pyrvinium reduced the $CD15^+$ cell population (**Figure 4D**). Like pyrvinium, PKF115-584 reduced $CD15^+$ cell numbers, whereas vismodegib had no effect (**Figure 4E**). Together, these results suggest that pyrvinium targets a WNT-driven $CD15^+$ cell population critical for self-renewal.

A $Sox2^+/CD15^+$ population was previously mentioned in the literature(28), but its regulation remains unclear. Given its ability to deplete both $Sox2^+$ and $CD15^+$ cells, we asked whether the $CD15^+$ cells targeted by pyrvinium also express Sox2. Flow cytometry analyses showed that most $CD15^+$ cells in MPC cultures are $Sox2^+$, and pyrvinium reduced the number of these double-positive cells (**Figure 4F**). WNT inhibition with PKF115-584 similarly reduced $Sox2^+$ cells (**Figure 4G**) and $Sox2^+/CD15^+$ cells (**Figure 4H**), whereas vismodegib did not (**Figure 4G, H**). In contrast, Gli inhibition by targeting BET reduced $Sox2^+$ cells, but did not affect $CD15^+$ or $Sox2^+/CD15^+$ cells (**Figure 4I**). Together, these findings suggest that pyrvinium depletes Smo inhibitor-resistant, Gli1-driven $Sox2^+$ cells, including a WNT-driven $CD15^+$ subset (**Figure 4J**).

CK1α agonists reduce MB stemness

Given pyrvinium's ability to deplete $Sox2^+$ and $CD15^+$ cells in MPC cultures, we tested whether it does the same in vivo. Mice were implanted subcutaneously with *Ptch1-LacZ*, *Trp53-KO* MB cells, and allowed to form tumors. Because of its poor systemic bioavailability(29), pyrvinium was administered subcutaneously near the tumor site(13). This local dosing suppressed tumor growth (**Figure 5A**; **Supplemental Figure 2A**), reduced proliferative index, and decreased the number of $Gli1^+$ and $Sox2^+$ cells, including double-positive ones (**Figure 5B**; **Supplemental Figure 2B**). Pyrvinium also depleted $CD15^+$ and β-$catenin^+$ cells, including a WNT-labeled $CD15^+$ subset, and diminished the $Sox2^+/CD15^+$ population within these tumors (**Figure 5B**; **Supplemental Figure 2B**).

Pyrvinium's poor bioavailability limits its use to subcutaneous tumor models(29). To test whether CK1α agonists reduce MB stemness in orthotopic MB, we used a brain-permeable derivative of pyrvinium, SSTC3(14, 17). Like pyrvinium, SSTC3 suppressed SHH and WNT target gene expression (**Figure 5C**), and reduced cell viability (**Figure 5D**) and self-renewal (**Figure 5E**) in MPC cultures. In mice bearing orthotopic *Ptch1-LacZ, Trp53-KO* tumors, SSTC3 reduced the proliferation index and the numbers of cells labeled for SHH as well as for WNT markers (**Figure 5F**; **Supplemental Figure 3A**). SSTC3 also reduced overall $Sox2^+$ cells, including $Gli1^+$ ones, and decreased $CD15^+$, $CD15^+$/β-catenin$^+$, as well as $Sox2^+/CD15^+$ populations (**Figure 5F**; **Supplemental Figure 3A**). To determine whether the efficacy of CK1α agonists goes beyond murine models, we tested SSTC3 in a *TP53*-mutant SHH PDOX model. While we previously showed that SSTC3 debulks *TP53*-mutant SHH tumors and prolongs survival in mice harboring these xenografts(14), we had not evaluated its effects on stemness markers. Here, we show that SSTC3 reduces $Sox2^+$, $CD15^+$, and $Sox2^+/CD15^+$ populations even in PDOX tumors (**Figure 5G**; **Supplemental Figure 3B**), despite a higher baseline of $Sox2^+$ cells compared to mouse-derived models, supporting efficacy across systems.

Pyrvinium reduces MB relapse risk

Based on their ability to reduce MB stemness and ex vivo self-renewal, we next examined whether CK1α agonists would also impair tumor-propagating potential using relapse-risk protocols in which primary sphere formation is assessed along with secondary tumor engraftment and time to engraftment (**Figure 6A**). Due to its FDA approval and ongoing cancer clinical trials (NCT05055323, NCT06782048, NCT06590454), we focused these studies on pyrvinium. In primary sphere formation assays, in which equal numbers of viable cells were plated and allowed to form spheres, results showed that pyrvinium-treated tumors fail to propagate in culture (**Figure 6B**). Similarly, limiting dilution assays in which equal numbers of viable tumor cells were subcutaneously re-implanted showed differences in engraftment ability between vehicle and pyrvinium-treated tumors (**Figure 6C**). Moreover, orthotopic re-engraftment showed that tumors derived from pyrvinium-treated residual disease took longer to establish (**Figure 6D**). These results show pyrvinium's potential to reduce SHHα MB relapse risk.

DISCUSSION

Here, we demonstrate that compounds activating CK1α attenuate MB stemness and, consequently, reduce relapse risk in animal models. Our findings suggest that these drugs function as dual inhibitors, simultaneously blocking Gli and WNT signaling, key drivers of $Sox2^+$ cell proliferation and $CD15^+$ self-renewal, respectively. The ability of CK1α agonists to attenuate MB stemness is particularly critical in SHHα MB, which accounts for ~75% of SHH MB recurrences(3), and shows 0-41% five-year-survival after relapse(3, 5). While we cannot exclude the possibility that pyrvinium and SSTC3 also attenuate stemness in *TP53*-wild-type SHH MB, this study was focused on the mutant subtype due to the clinical need they represent.

We previously showed that $Sox2^+$ MB cells depend on the activation of Gli downstream of Smo, and that these cells are susceptible to depletion by BET inhibitors, which block Gli transcriptional activity(30, 31). However, efforts to translate these findings have been limited by the clinical toxicity of BET inhibitors(32) and the recent termination of a pediatric brain tumor trial testing them (NCT03936465). Dose-limiting toxicities of BET inhibitors underscores the need for alternative approaches to suppress Gli transcriptional activity. Because Gli proteins are transcription factors and therefore difficult to target directly(33), we focused instead on regulatory mechanisms controlling Gli stability. Since CK1α phosphorylates Gli, priming it for degradation(11), we tested pyrvinium, a previously characterized CK1α agonist(12), in MB cultures enriched for $Sox2^+$ cells(9). Pyrvinium suppressed SHH signaling and reduced the proliferation of $Sox2^+$ cells. Thus, this drug shows efficacy like that of BET inhibitors, but with the potential advantage of lower toxicity.

Besides breast cancer(34), CK1α agonists have been reported to attenuate glioblastoma stemness by targeting $CD133^+$ cells(35). Although early work described $CD133^+$ MB cells as the self-renewing pool(36), subsequent studies pointed to $CD15^+$ cells instead (24-26). Thus, based on our data showing that pyrvinium blocks MB self-renewal, we studied whether pyrvinium targets $CD15^+$ cells. Here, we showed that pyrvinium depletes a self-renewing $CD15^+$ cell pool, and it does so by targeting WNT signaling. This aligns with previous evidence implicating WNT signaling in regulating cancer stemness(37). However, while previous works implicated CD15 cells with WNT signaling in retina progenitors(38), to our knowledge, this is the first study to link $CD15^+$ cells with WNT signaling in cancer. Mechanistically, we found that, similar to previous observations(22), WNT activation in SHHα MPC cultures is associated with the loss of p53*/miR-34a*-mediated repression of WNT components such as LEF1. While additional studies are needed to clarify why WNT signaling is specifically required in $CD15^+$ cells or which other WNT signaling components are repressed by *miR-34a* in MB, our findings identified a connection between p53 loss, WNT signaling, and CD15-driven MB self-renewal.

Our data support that pyrvinium exerts its effects on stemness by inhibiting Gli and WNT signaling. However, pyrvinium has been reported to inhibit AKT signaling(39), a pathway known to promote survival and maintenance of tumor stem-like cells(40), including those in MB(41). Similarly, pyrvinium can attenuate STAT3 activity(42, 43), which is a critical regulator of stemness and self-renewal in multiple cancers(44), including MB(45). Thus, the anti-stemness effects of pyrvinium may involve the suppression of a broader group of oncogenic signaling programs that converge on regulating stemness. Additionally, pyrvinium has been shown to preferentially accumulate in the mitochondria of tumor cells(46), where it disrupts mitochondrial respiration(29, 47) and impairs mitochondrial gene expression(48). Notably, tumor stem-like cells exhibit heightened mitochondrial requirements(49), providing an additional basis for the preferential targeting of these populations by pyrvinium.

While our work primarily focused on pyrvinium due to its FDA approval and its use in recently launched trials for pancreatic and gastric cancer (NCT05055323, NCT06782048, NCT06590454), it is important to note that pyrvinium does not cross the blood-brain barrier. Although our ex vivo pyrvinium concentrations are comparable to those used in other tumor models(13, 16, 34, 35, 39, 42, 43, 46, 48, 50), including those supporting its translation(48, 51), its lack of brain permeability prevents direct in vivo dose comparisons in MB and limits its direct translational potential for these tumors. While strategies such as intrathecal drug administration could be considered, a more promising approach is the development of derivatives with improved brain permeability. SSTC3, evaluated here in two different orthotopic SHHα MB models, serves as a useful brain-penetrant tool compound. SSTC3 reduced stemness markers, and as shown in our previous work also prolonged survival in SHH MB orthotopic models, including in a *TP53*-mutant PDOX(14). Although we initially attributed the survival advantage of SSTC3-treated mice to tumor debulking, our new data suggest that it may in addition result from its effects on tumor stemness. Together, these findings suggest that the efficacy of CK1α agonists extends to stemness regulation and supports their further development, particularly brain-permeable derivatives, for very high-risk SHHα MB patients.

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AUTHOR CONTRIBUTIONS

K.P.: Conceptualization, methodology, investigation, visualization, and manuscript writing.

M.T.-C.: Methodology, investigation, animal studies, and visualization.

A.D.S.: Methodology, investigation, animal studies, and visualization.

P.S.: Methodology, investigation, and visualization.

M.S.-S.: Methodology, investigation, animal studies, and visualization.

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D.T.W.: Conceptualization, methodology, investigation, and visualization.

D.L.F.: Conceptualization.

V.M.: Conceptualization, visualization, and manuscript writing.

T.B.: Conceptualization, visualization, and manuscript writing.

J.R.-B.: Conceptualization, methodology, investigation, visualization, and manuscript writing.

CONFLICTS OF INTEREST

D.L.F. is a full-time employee at StemSynergy Therapeutics Inc. The rest of the authors declare no conflicts of interest.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

No human subjects were involved in this study. All animal experiments were conducted in compliance with the National Institutes of Health (NIH) Guide for the Care and Use of Laboratory Animals. Experimental protocols were reviewed and approved by the Institutional Animal Care and Use Committees (IACUC) at the Medical University of South Carolina (IACUC-2019-00870-1) and the University of Miami (13-217).

DATA AVAILABILITY

All data presented in this manuscript will be made available upon request. Uncropped versions of all immunoblotting images in this manuscript can be found in Supplemental Material.

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FIGURE LEGENDS

Figure 1. Pyrvinium attenuates SHH signaling in MPC cultures. **A**. SHH signaling was induced in *Gli1* reporter cells with 100 nM SAG for 24h, followed by treatment with the indicated concentrations of pyrvinium for an additional 24h. **B**. Organotypic cerebellar slices from 7-day-old mice were treated with 200 nM pyrvinium for 24h, followed by a BrdU pulse. Quantification of $BrdU^+$ cells is shown. Data were analyzed using an unpaired t-test. Calbindin was used to label the Purkinje layer. **C**. Gli1 levels were quantified by immunoblotting in MPC-2 cultures treated with increasing concentrations of pyrvinium for 24h (OD: optical density). **D**. Gli1 levels were similarly evaluated in the indicated MPC cultures treated with 200 nM pyrvinium for 24h. **E**. MPC-1 cultures were treated with the indicated concentrations of pyrvinium for 24h, and Gli1-labeled cells were quantified by flow cytometry. **F**. Expression of SHH target genes was measured by RT-qPCR in MPC-2 cultures treated with increasing concentrations of pyrvinium for 24h. **G**. MPC cultures were treated with pyrvinium for 72h, and cell viability was assessed using an MTT assay. **H**. MPC-2 cultures were exposed to 200 nM pyrvinium for 24h before assessing BrdU incorporation and cleaved Caspase-3^+ (C-Casp3). Quantification of positive cells is shown. Data

was analyzed using an unpaired t-test. **I**. Levels of the indicated proteins were quantified in extracts from MPC-2 cultures treated with increasing concentrations of pyrvinium for 24h. **J**. Levels of the indicated proteins were assessed in extracts from the indicated MPC cultures treated with 200 nM pyrvinium for 24h. **K**. MPC-1 cultures were treated with the indicated concentrations of pyrvinium for 24h, and Ki67-labeled cells were quantified by flow cytometry. **L**. MPC-1 cultures were exposed to the indicated concentrations of pyrvinium, and numbers of $Sox2^+$ cells were determined by flow cytometry. **M**. MPC-1 cultures were exposed to the indicated concentrations of pyrvinium, and numbers of $Sox2^+/Gli1^+$ and $Sox2^+/Ki67^+$ cells were determined by flow cytometry. In all cases, representative IF images (scale bar 100 µm) and flow cytometry plots are shown. Unless otherwise indicated, mean ± SEM of data normalized to DMSO were analyzed using a one-way ANOVA followed by Dunnett's post-hoc. All EC_{50}s were calculated using non-linear regression analyses. * $p< 0.05$, ** $p< 0.01$, *** $p< 0.001$, **** $p< 0.0001$.

Figure 2: Pyrvinium attenuates MB self-renewal. A. Schematic of the self-renewal protocol. **B.** MPC-2 cultures were exposed to increasing concentrations of indicated BET inhibitors for 24h before allowing equal numbers of viable cells to form new spheres for one week. **C.** Similar cultures were exposed to increasing concentrations of vismodegib before completing self-renewal protocols. **D.** MPC-2 cultures were transfected with siRNA pools targeting *Gli1* and *Gli2*, as well as scramble *siRNA* (*siSC*) and GFP-labeled controls. Seventy-two hours after transfection, spheres were dissociated, and equal numbers of viable cells were replated. A one-way ANOVA was used to analyze data normalized to *siSC*. **E.** MPC cultures were allowed to form secondary spheres following a 24h incubation with the indicated concentrations of pyrvinium. **F.** Indicated numbers of viable cells from MPC-2 cultures treated with 200 nM pyrvinium for 24h were subcutaneously implanted. The frequency of tumor engraftment was analyzed using a one-sided χ^2 test. In all cases,

representative images of self-renewal assays are shown. Unless otherwise indicated, mean ± SEM of data normalized to DMSO were analyzed using non-linear regression tests. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Figure 3: Pyrvinium acts on WNT signaling to block MB self-renewal. A. WNT signaling was induced in TCF/LEF1 reporter cells by 100 ng/ml WNT3a and 10 mM LiCl for 24h before treatment with pyrvinium for an additional 24h. Data was analyzed using a one-way ANOVA followed by Dunnett's post-hoc. **B**. The levels of β-catenin following 24h incubation with increasing concentrations of pyrvinium were quantified in MPC-2 cultures by immunoblotting. (OD: optical density). **C**. The levels of β-catenin following 24h incubation with 200 nM pyrvinium were similarly determined in the indicated MPC cultures. **D**. MPC-2 cultures were exposed to indicated concentrations of pyrvinium for 24h before determining the expression of indicated WNT target genes by RT-qPCR. **E.** MPC-2 cultures were transfected with the indicated vectors and expression of WNT target genes was determined 48h later. Data was normalized to *pcDNA* control. **F.** MPC-2 cultures were transfected with the indicated vectors. 3 days post-transfection cells were treated with 200 nM pyrvinium for an additional 24h before sphere dissociation and re-plating of equal numbers of viable cells. Numbers of secondary spheres were quantified 7 days later. Data was normalized to *pcDNA* control. **G**. The expression of *miR-34a-5p* was determined in MPC-2 (*Trp53*$^{-/-}$) and MPC-47 (*Trp53*$^{+/+}$) *Ptch1-LacZ* cultures by RT-qPCR and normalized to MPC-2 data. **H**. MPC-2 cultures were transfected with specific *mimic*- or *anti-miR* sequences along with respective scramble (*miR-SC*) controls, and LEF1 levels were quantified by immunoblotting 48h later. **I**. MPC-47 cells were transfected with *anti-miR-34a* or a scramble control, and the expression of WNT target genes was determined 48h later by RT-qPCR. Results were normalized to the corresponding scramble sequence. **J**. MPC-2 cultures were transfected with

indicated *mimic*- and *anti-miR* sequences, and secondary sphere formation was determined 72h later. Results were normalized to the corresponding scramble sequence. **K**. Schematic of the mechanism of activation of WNT signaling in *Trp53* mutant MPC cultures. In all cases, representative images of self-renewal assays are shown. Unless otherwise indicated, mean ± SEM data were analyzed using an unpaired t-test. All EC_{50}s were calculated using non-linear regression analyses. * $p< 0.05$, ** $p< 0.01$, *** $p< 0.001$, **** $p< 0.0001$.

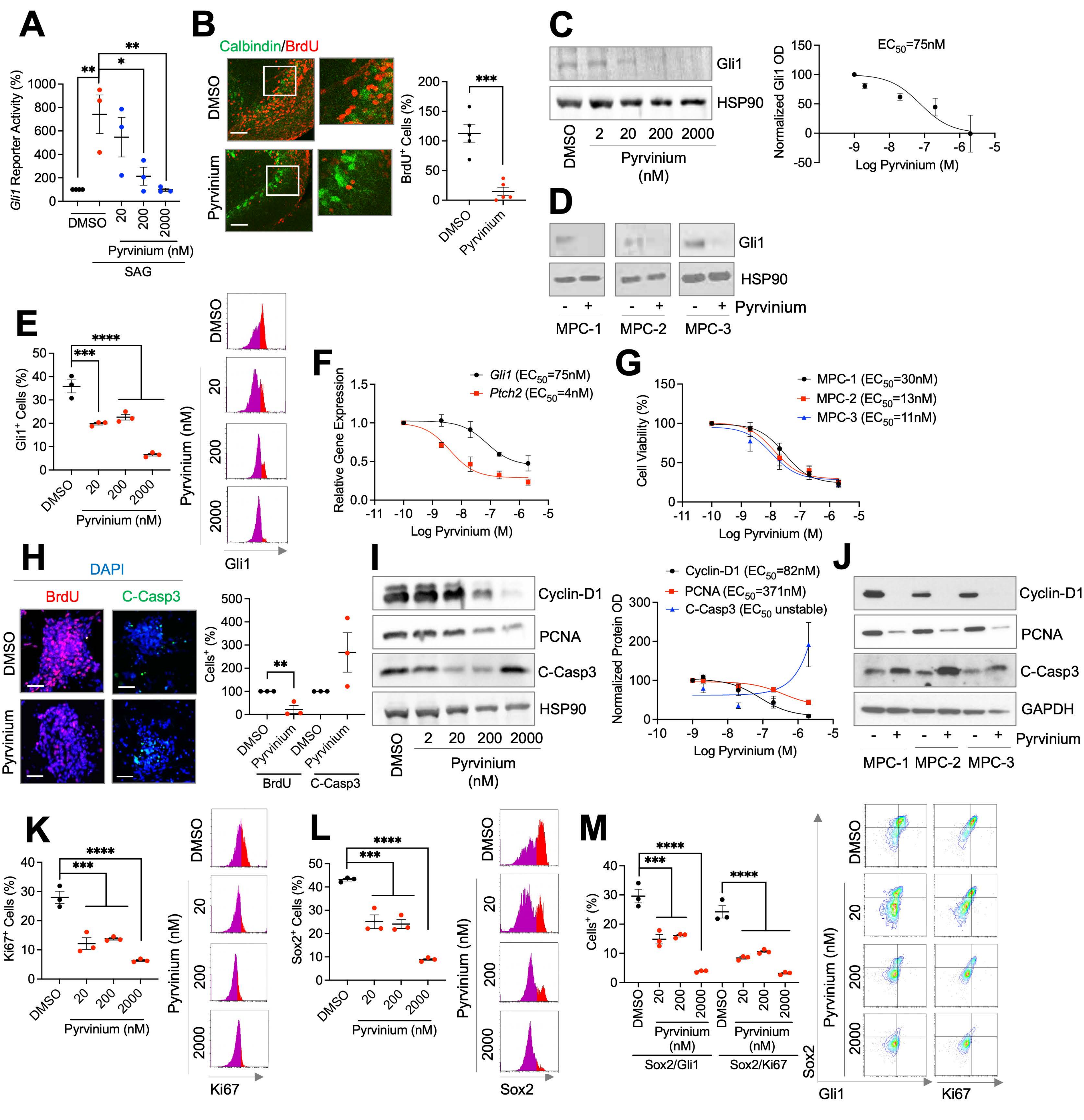
Figure 4: Pyrvinium depletes a self-renewing CD15^{+} cell population. **A.** CD15^{+}/Ter-119^{-} cells were flow-sorted from *Ptch1-LacZ, Trp53-KO* MB tumors. **B.** Similarly sorted cells were plated in the presence of vismodegib (200 nM), PKF115-584 (1000 nM), or pyrvinium (200 nM), and their ability to form spheres was determined. **C.** RNA was extracted from CD15^{+}/Ter-119^{-} and CD15^{-}/Ter-119^{-} sorted pools, and expression of WNT and SHH target genes was determined by RT-qPCR. Data were normalized to CD15^{-} cells and analyzed using an unpaired t-test. **D.** MPC-1 cultures were exposed for 24h to increasing concentrations of pyrvinium and numbers of CD15^{+} cells were determined by flow analysis. **E.** Similar cultures were exposed to vismodegib (200 nM) or PKF115-584 (1000 nM) for 24h, and CD15^{+} cell numbers were determined by flow analysis. **F**. The number of Sox2^{+}/CD15^{+} cells was determined by flow cytometry in MPC-1 cultures treated with increasing concentrations of pyrvinium for 24h. **G**. Similar cultures were exposed to vismodegib (200 nM) or PKF115-584 (1000 nM) for 24h, and Sox2^{+} cell numbers were determined by flow analysis. **H**. The number of Sox2^{+}/CD15^{+} cells was determined by flow cytometry in similarly treated cultures. **I**. MPC-1 cultures were exposed to I-BET151 (500 nM) for 24h, and the numbers of indicated positive cells were determined by flow analysis. Data was analyzed using an unpaired t-test. **J.** A schematic of the mechanism of action of pyrvinium in MPC cultures. Unless

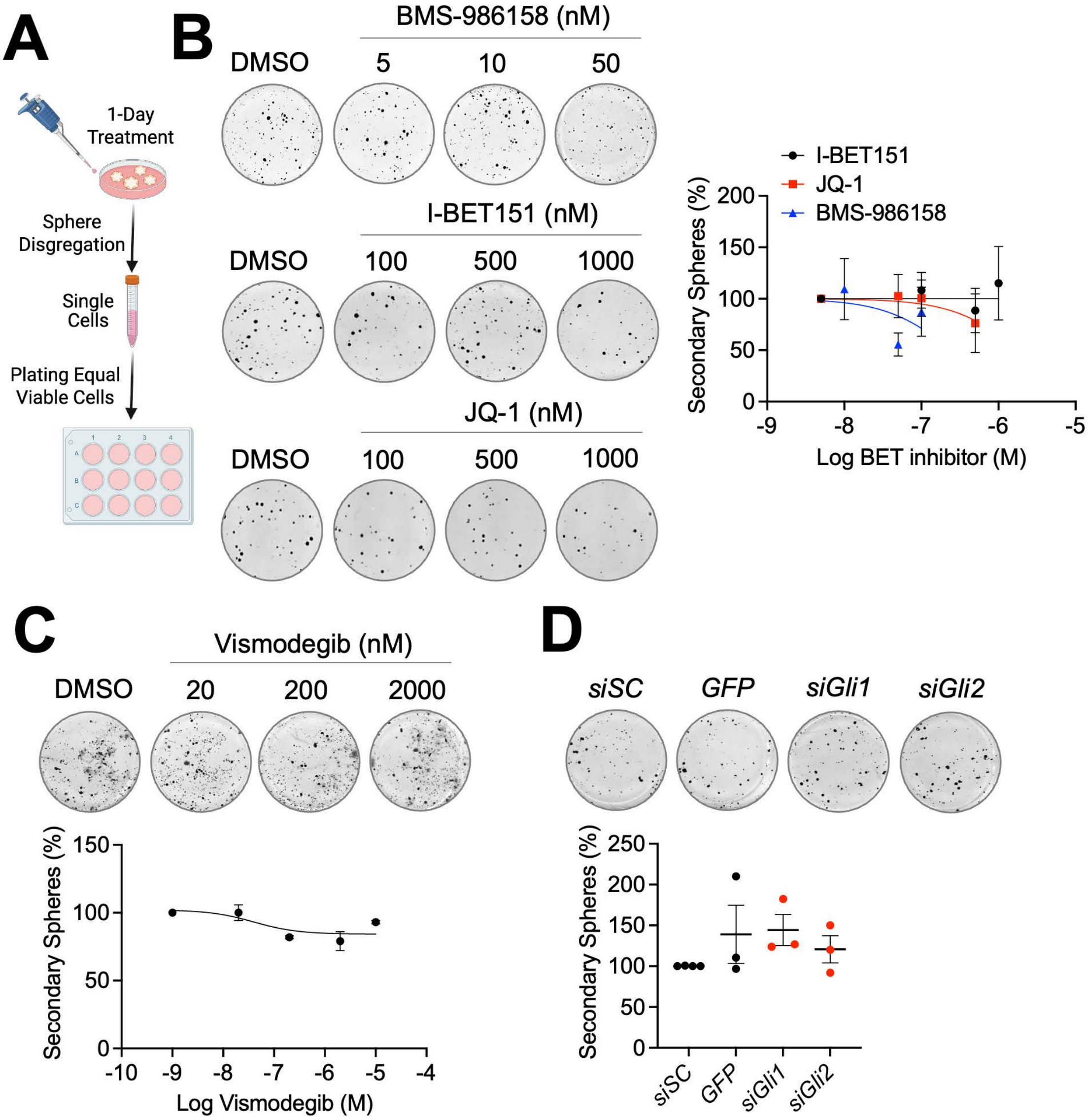
otherwise indicated, mean ± SEM of data normalized to DMSO were analyzed using a one-way ANOVA followed by Dunnett's post-hoc. * $p< 0.05$, ** $p< 0.01$, *** $p< 0.001$, **** $p< 0.0001$.

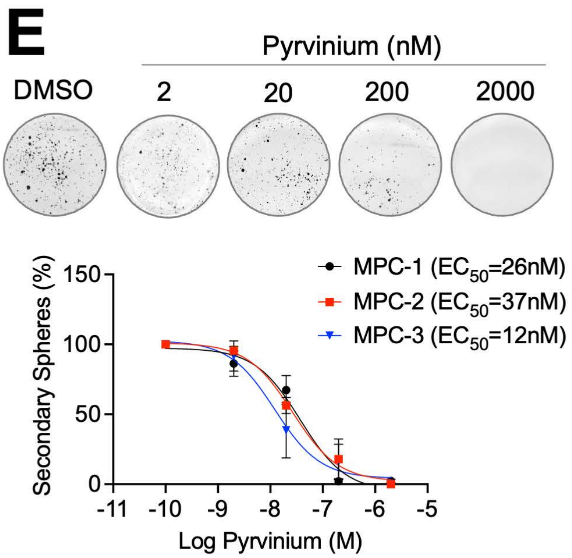
Figure 5: CK1α agonists reduce MB stemness. A. *Ptch1-LacZ, Trp53-KO* MB cells were subcutaneously implanted. Once tumors reached ~200 mm^3, mice were treated with pyrvinium (0.8 mg/kg, s.c., q.o.d.) or vehicle, and tumor volumes were measured over time. Representative H&E staining (scale bar 500 µm) at day 7 is shown. **B.** Mice harboring similar tumors received three doses of pyrvinium, and cells positive for the indicated markers were quantified by flow analysis. **C**. MPC-2 cultures treated with 1000 nM SSTC3 for 24h were analyzed by RT-qPCR for SHH and WNT target gene expression. **D**. Similar cultures were exposed to increasing SSTC3 concentrations, and cell viability was measured by MTT assay 72h later. **E**. Similar cultures were allowed to form secondary spheres after 24h SSTC3 treatment. **F**. Mice were orthotopically implanted with *Ptch1-LacZ; Trp53-KO* MB cells, then treated with SSTC3 (10 mg/kg, i.p., q.d.) for 3 days starting 10 days post-surgery. Brains were harvested, and numbers of cells positive for the indicated markers were quantified by flow cytometry. **G**. Mice with orthotopic SJSHHMB-14-7196 tumors (grown for 30 days) were treated with SSTC3 (10 mg/kg, i.p., q.d.) for 3 days before IHC staining for Sox2 and CD15. Representative images (scale bar 20 µm) are shown. Unless noted, mean ± SEM of data normalized to DMSO were analyzed using an unpaired t-test. All EC_{50}s were calculated using non-linear regression analyses. *$p< 0.05$, **$p< 0.01$, ***$p< 0.001$, ****$p< 0.0001$.

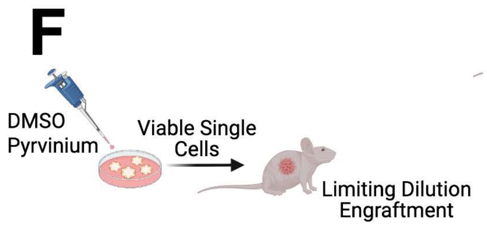
Figure 6: Pyrvinium reduces MB relapse risk. A. Schematic of relapse-risk protocols: mice bearing subcutaneous *Ptch1-LacZ, Trp53-KO* MB received 3 doses of pyrvinium before preparing single-cell suspensions for primary sphere assays or subcutaneous and orthotopic engraftment. **B.** Quantification and representative images of primary spheres from vehicle- and pyrvinium-treated

tumors are shown. The data were normalized to one vehicle-treated animal and analyzed using an unpaired t-test. **C.** Tumor engraftment frequency of limiting cell numbers was analyzed with a one-sided χ^2 test. **D.** Time to relapse was assessed by symptom-free survival and analyzed by Log-rank (Mantel-Cox) tests. Unless noted, mean ± SEM of data. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$.

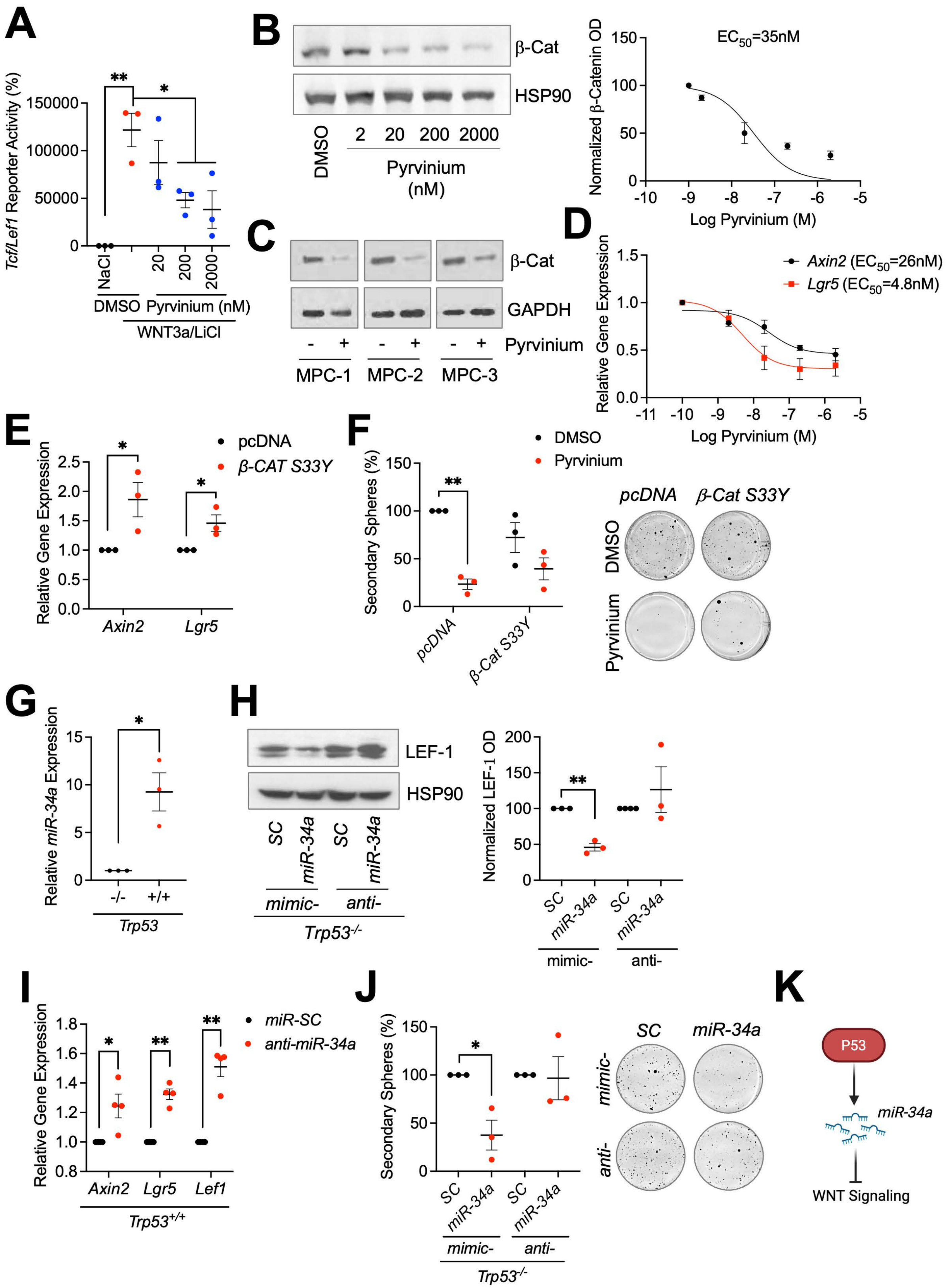

A
Gli1 Reporter Activity (%)
DMSO
20
200
2000
Pyrvinium (nM)
SAG
B
Calbindin/BrdU
DMSO
Pyrvinium
BrdU+ Cells (%)
C
Gli1
HSP90
DMSO
2 20 200 2000
Pyrvinium (nM)
EC50=75nM
Normalized Gli1 OD
Log Pyrvinium (M)
D
- + Pyrvinium
MPC-1 MPC-2 MPC-3
E
Gli1+ Cells (%)
Pyrvinium (nM)
Gli1
F
Gli1 (EC50=75nM)
Ptch2 (EC50=4nM)
Relative Gene Expression
Log Pyrvinium (M)
G
MPC-1 (EC50=30nM)
MPC-2 (EC50=13nM)
MPC-3 (EC50=11nM)
Cell Viability (%)
Log Pyrvinium (M)
H
DAPI
BrdU
C-Casp3
Cells+ (%)
I
Cyclin-D1
PCNA
C-Casp3
HSP90
Cyclin-D1 (EC50=82nM)
PCNA (EC50=371nM)
C-Casp3 (EC50 unstable)
Normalized Protein OD
J
GAPDH
K
Ki67+ Cells (%)
Ki67
L
Sox2+ Cells (%)
Sox2
M
Sox2/Gli1
Sox2/Ki67
Gli1
Ki67







Cell (#)	**DMSO** Engrafted Cells	**Pyrvinium** Engrafted Cells
1×10^6	2/2	0/2
5×10^5	7/8	0/8*
1.5×10^5	2/3	0/3



A
Tcf/Lef1 Reporter Activity (%)
150000
100000
50000
0
**
*
NaCl
20
200
2000
DMSO
Pyrvinium (nM)
WNT3a/LiCl
B
β-Cat
HSP90
DMSO
2
20
200
2000
Pyrvinium (nM)
Normalized β-Catenin OD
EC50=35nM
Log Pyrvinium (M)
C
β-Cat
GAPDH
Pyrvinium
MPC-1
MPC-2
MPC-3
D
Relative Gene Expression
Axin2 (EC50=26nM)
Lgr5 (EC50=4.8nM)
Log Pyrvinium (M)
E
Relative Gene Expression
pcDNA
β-CAT S33Y
Axin2
Lgr5
F
Secondary Spheres (%)
DMSO
Pyrvinium
pcDNA
β-Cat S33Y
G
Relative miR-34a Expression
-/-
+/+
Trp53
H
LEF-1
HSP90
SC
miR-34a
mimic-
anti-
Trp53-/-
Normalized LEF-1 OD
I
Relative Gene Expression
miR-SC
anti-miR-34a
Axin2
Lgr5
Lef1
Trp53+/+
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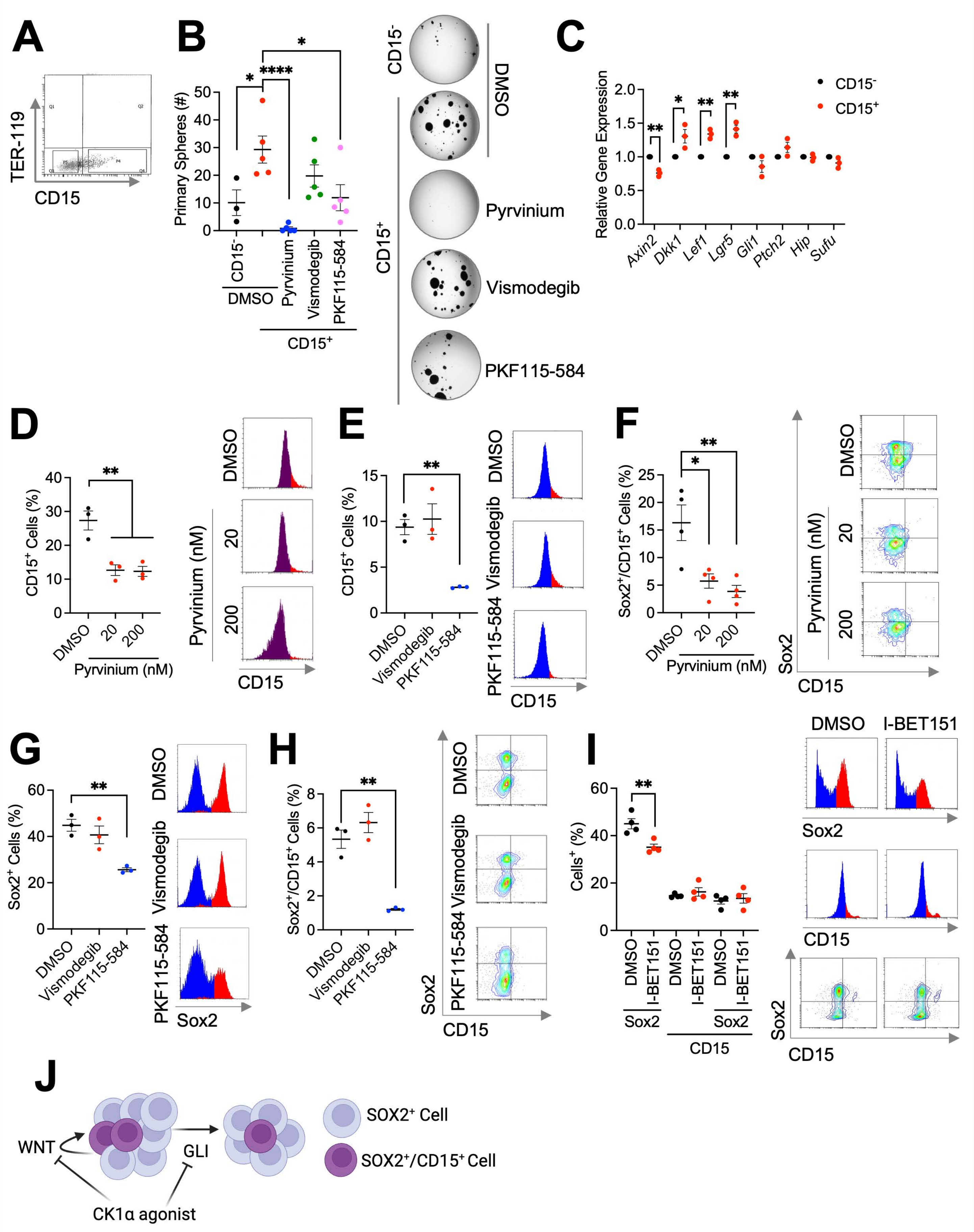
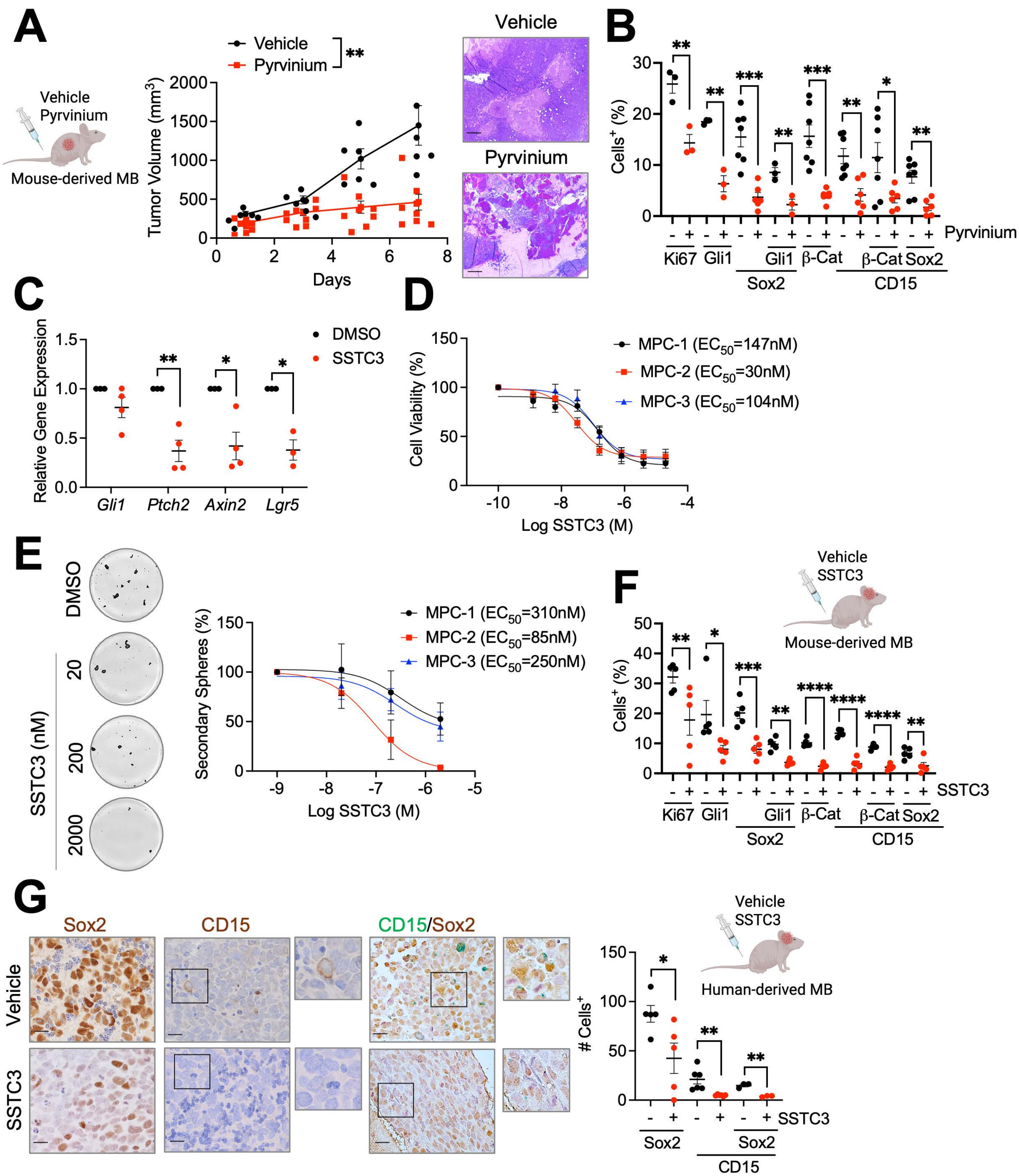
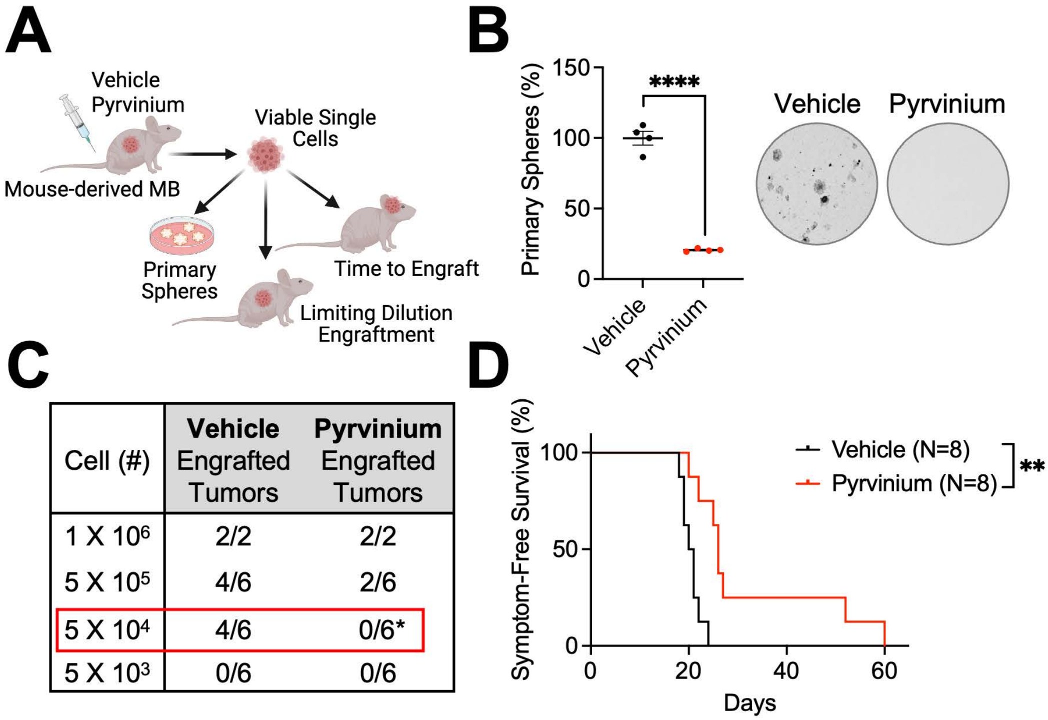
P53
miR-34a
WNT Signaling



A
TER-119
CD15
B
Primary Spheres (#)
CD15-
DMSO
Pyrvinium
Vismodegib
PKF115-584
CD15+
C
Relative Gene Expression
Axin2
Dkk1
Lef1
Lgr5
Gli1
Ptch2
Hip
Sufu
CD15-
CD15+
D
CD15+ Cells (%)
DMSO
Pyrvinium (nM)
CD15
E
F
Sox2+/CD15+ Cells (%)
Sox2
G
Sox2+ Cells (%)
H
I
Cells+ (%)
I-BET151
J
WNT
GLI
CK1α agonist
SOX2+ Cell
SOX2+/CD15+ Cell


A
Vehicle
Pyrvinium
Mouse-derived MB
Vehicle
Pyrvinium
**
Tumor Volume (mm3)
Days
B
Cells+ (%)
Pyrvinium
Ki67
Gli1
Sox2
β-Cat
CD15
C
DMSO
SSTC3
Relative Gene Expression
Gli1
Ptch2
Axin2
Lgr5
D
MPC-1 (EC50=147nM)
MPC-2 (EC50=30nM)
MPC-3 (EC50=104nM)
Cell Viability (%)
Log SSTC3 (M)
E
DMSO
SSTC3 (nM)
20
200
2000
MPC-1 (EC50=310nM)
MPC-2 (EC50=85nM)
MPC-3 (EC50=250nM)
Secondary Spheres (%)
Log SSTC3 (M)
F
Vehicle
SSTC3
Mouse-derived MB
Cells+ (%)
G
Sox2
CD15
CD15/Sox2
Vehicle
SSTC3
Human-derived MB
Cells+


A
Vehicle
Pyrvinium
Mouse-derived MB
Viable Single Cells
Primary Spheres
Time to Engraft
Limiting Dilution Engraftment
B
Primary Spheres (%)
150
100
50
0
****
Vehicle
Pyrvinium
Vehicle
Pyrvinium
C
Cell (#)
Vehicle Engrafted Tumors
Pyrvinium Engrafted Tumors
1 X 10⁶ 2/2 2/2
5 X 10⁵ 4/6 2/6
5 X 10⁴ 4/6 0/6*
5 X 10³ 0/6 0/6
D
Symptom-Free Survival (%)
100
50
0
0
20
40
60
Days
Vehicle (N=8)
Pyrvinium (N=8)
**